

Moussa Atwi

PhD Researcher · Scientific Computing & Machine Learning

Open to industry roles in mathematical modeling, scientific computing, and machine learning.

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📍 Aachen, Germany

[in](#) [Musa Atwi](#)

[AtwiMusa](#)

[Portfolio](#)

EXPERIENCE

- **PhD Researcher** Full-time · 30 h/wk
RWTH Aachen University · Germany Aug 2023 – Present
"Sparse Exit Wave Reconstruction via Deep Unfolding"
Incorporating data-driven learning into model-based exit wave reconstruction, a variant of phase retrieval arising in electron microscopy, by unfolding classical optimization algorithms into trainable neural networks.
[Phase Retrieval](#) · [Image Processing](#) · [Optimization](#) · [Deep Neural Networks](#)
- **Publication**
"A generalization bound for exit wave reconstruction via deep unfolding"
[Preprint](#) · [related to PhD](#) – [arXiv 2025](#)
- **E-Learning & Assessment Assistant** Mini-job · 10 h/wk
RWTH Aachen University · Germany Apr 2026 – Present
Managing exercises and online assessments via Dynexite, coordinating Bonus Points Tests, publishing materials on Moodle, and grading student work.
[Dynexite](#) · [Moodle](#) · [Online Assessment](#) · [Grading](#)
- **Research Internship & Master's Thesis** Part-time · 20 h/wk
Zuse Institute Berlin · for the M.Sc. at TU Berlin Sep 2022 – May 2023
"Approximation of Koopman Operators Using Random Basis Functions"
Studied Koopman operator theory and nonlinear dynamical systems, including Koopman eigenfunctions, invariant subspaces, and the Koopman embedding. Implemented Extended Dynamic Mode Decomposition combined with monomials, random Gaussians, and trained and untrained neural networks in Python.
[Koopman Operators](#) · [Neural Networks](#) · [Dynamical Systems](#)
- **Research Internship & Master's Thesis** Full-time · 35 h/wk
Université de Bordeaux · for the M.Sc. at UPPA, France Feb – Aug 2020
"Semi-discrete Optimal Transport for Large-Scale Problems"
Studied the Monge and Kantorovich problems, the dual formulation, the c-transform, and Laguerre cells. Implemented gradient descent algorithms in Python to solve large-scale semi-discrete optimal transport problems in image processing and machine learning, analysing convergence behaviour.
[Convex Duality](#) · [Image Modeling](#) · [Gradient Descent](#)

- **Master's Thesis — Control Theory**
Lebanese International University · Lebanon Feb – Aug 2019
"Decay Rates of a Timoshenko System with Kelvin-Voigt Damping"
Proved that the energy of the Timoshenko system with Kelvin-Voigt damping dissipates exponentially or polynomially, with the decay rate depending on the properties of the material coefficient function.
[C0-Semigroup](#) · [Timoshenko Beam](#) · [Polynomial & Exponential Stability](#)

- **Mathematics Teacher**
Private Schools · Beirut, Lebanon 2012 – 2018
Worked four years as a school teacher of mathematics in English at private high schools, covering intermediate and secondary levels from Grades 6 to 10. Alongside school teaching, provided private tutoring to university students in Real Analysis and Algebra, and to high-school students in general mathematics.
[Mathematics Teaching](#) · [Secondary Education](#) · [Private Tutoring](#)

PROJECT · FOR THE M.SC. AT LIU, LEBANON

- **Project — Finite Element Methods**
"Piecewise Linear Finite Element Methods for a 1D Elliptic Equation"
Estimated the order of convergence in Matlab on uniform and non-uniform meshes for a one-dimensional elliptic equation.
[Finite Element Method](#) · [Convergence Analysis](#) · [Matlab](#)

EDUCATION

- **M.Sc. Scientific Computing**
Technische Universität Berlin 2021 – 2023
GPA 1.2 · Excellent
- **M.Sc. Mathematics, Modeling & Simulation**
Université de Pau et des Pays de l'Adour 2019 – 2020
Excellence Scholarship · Région Nouvelle-Aquitaine
- **M.Sc. Applied Mathematics**
Lebanese International University 2018 – 2019
Rank 1 · GPA 4.0 · Excellent
- **Master 1, Pure Mathematics** (1st Year of Master)
Lebanese University 2014 – 2015
Ranked 1st in Number Theory
- **B.Sc. Pure Mathematics**
Lebanese University 2008 – 2012

ACHIEVEMENTS

- Received **A** in the research internship, master's thesis, and most courses of the M.Sc. at TU Berlin.
- Received **A** in all required courses of the M.Sc. at LIU.
- **Excellence Scholarship** from Région Nouvelle-Aquitaine to pursue the master's at the University of Pau.
- **Ranked 1st** in the Number Theory course, Master 1 at LU.
- Published Arabic poet — **Rashfato Alami**, a collection of 80+ romantic poems.

SKILLS

Scientific & Mathematical

- Linear Optimization
- Convex Optimization
- Optimal Control
- Proximal Gradient
- Machine & Deep Learning
- Deep Unfolding
- Phase Retrieval
- Numerical Simulation
- Runge-Kutta Methods
- Numerical Analysis
- Nonlinear Optimization
- Non-Convex Optimization
- Optimal Transport
- Calculus of Variations
- Neural Networks
- Bayesian Inference
- Koopman Theory
- FDM / FEM / FVM
- Multi-step Methods
- Partial Differential Equations

Computing

- Python
- PyTorch
- Deeptime
- DifferentialEquations.jl
- Matlab
- DuMuX
- elsA-ONERA
- Linux · VS Code
- NumPy · Matplotlib
- TensorFlow
- Julia
- ExtendableGrids.jl
- C/C++ · OpenMP/MPI
- ANSYS-FLUENT
- Git · GitHub
- LaTeX · SPSS

Personal

- Strong Working Relationships
- Problem Solving
- Adaptability
- Teamwork & Collaboration
- Decision Making
- Fast Learner

- **Seminar — Optimal Transport**
"Entropic Regularization of Discrete and Continuous Optimal Transport"
Studied the Kantorovich and regularized problems, discrete and relative entropies, scaling through the Gibbs kernel, and Sinkhorn's algorithm for entropic optimal transport with quadratic cost. Derived the dual problem where strong duality holds.
[Sinkhorn's Algorithm](#) · [Dual Formulation](#) · [KL-Divergence](#) · [Gibbs Kernel](#)
- **Seminar — Nonlinear Optimization**
"Constrained Optimization and Duality — Applications to ML"
Studied primal and dual problems, constraint qualifications for strong duality, and norm-constrained optimization with barrier methods. Formulated the SVM dual, solved using coordinate and gradient descent methods.
[Lagrangian Duality](#) · [SVM Dual](#) · [Coordinate & Gradient Descent](#)
- **Project — Scientific Computing**
"Gas Transport in Porous Medium"
Numerical solution of the nonlinear Porous Medium Equation using the Voronoi Finite Volume Method in Julia. Applied Rothe's method, the implicit Euler method, and ExtendableGrids.jl for space and time discretization, with 1D and 2D simulations and performance improvements via DifferentialEquations.jl.
[Porous Medium Equation](#) · [Voronoi FVM](#) · [Julia](#)

- **Project — Porous Media Flow**
"Two-Phase Flow in Homogeneous Porous Media"
Summarized a paper on homogenization for reactive porous-media flow. Approximated the Buckley-Leverett equation for immiscible, incompressible two-phase flow in 1D and 2D homogeneous porous media using DuMuX, with analytic and random permeability.
[Buckley-Leverett Equation](#) · [DuMuX](#) · [Numerical Simulation](#)
- **Project — Fluid Mechanics**
"Numerical Simulation of Compressible and Incompressible Fluids"
Physical and mathematical modeling for numerical simulation. Used ANSYS-FLUENT for compressible and incompressible fluids and elsA-ONERA for compressible flows.
[ANSYS-FLUENT](#) · [elsA-ONERA](#) · [Compressible Flow](#)
- **Project — Wave Propagation**
"Inverse Problem for Propagation Media"
Implemented an inverse problem in Matlab to determine the nature of a propagation medium using a discontinuous Galerkin approximation.
[Inverse Problem](#) · [Discontinuous Galerkin](#) · [Matlab](#)

- **Grocery Delivery Rider** Mini-job · 10 h/wk
[Gorillas GmbH · Berlin, Germany](#) Oct 2021 — Apr 2022
Picked up grocery orders from the warehouse and delivered them to customers across Berlin by e-bike, ensuring timely service and customer satisfaction.
[Urban Mobility](#) · [E-bike Delivery](#)
- **COVID-19 Test Center** Mini-job · 10 h/wk
[Teststation Kantstrasse · Berlin](#) Jun — Aug 2021
Performed rapid antigen tests and guided customers through the registration and testing process at the Kantstrasse test station in Berlin.
[COVID-19 Testing](#) · [Customer Assistance](#)

- **Machine Learning Specialization** 2024
[Coursera · DeepLearning.AI & Stanford](#)
 - Supervised Learning: Regression & Classification
 - Advanced Learning Algorithms
 - Unsupervised Learning, Recommenders & Reinforcement Learning
- **Public Speaking & Presentation Skills** 2025
[RWTH Aachen University](#)
- **Unconscious Bias & Intercultural Training** 2025
[RWTH Aachen University](#)

Published Arabic poet — [Rashfato Alami](#), a first volume of 80+ romantic poems. Read the collection on the [portfolio](#).

Arabic Native **English** C1 **German** B1 **French** A1